Problem

Q. Suppose $x^h y^k$ is an IF for the DE

$$x^a y^b(my\,dx + nx\,dy) + x^c y^d(py\,dx + qx\,dy) = 0 \qquad (5)$$

Then find the value of h and k if $np \neq mq$.
Solution: Multiplying $x^h y^k$ with the given DE,

$$\left(x^{a+h}y^{b+k}\,my + x^{c+h}y^{d+k}\,py\right)dx + \left(x^{a+h}y^{b+k}\,nx + x^{c+h}y^{d+k}\,qx\right)dy =$$

which is exact. Here $M = m\,x^{a+h}y^{b+k+1} + p\,x^{c+h}y^{d+k+1}$ and $N = n\,x^{a+h+1}y^{b+k} + q\,x^{c+h+1}y^{d+k}$.
Using the condition $M_y = N_x$, we obtain

$$\begin{aligned} \Rightarrow \quad & m\,x^{a+h}(b+k+1)\,y^{b+k} + p\,x^{c+h}(d+k+1)\,y^{d+k} \\ & = n(a+h+1)\,x^{a+h}y^{b+k} + q(c+h+1)\,x^{c+h}y^{d+k} \end{aligned}$$



Solution...

Equating the coefficients,

$$m(b+k+1) = n(a+h+1)$$
$$p(d+k+1) = q(c+h+1)$$

$$\Rightarrow \quad nh - mk = m(b+1) - n(a+1)$$
$$qh - pk = p(d+1) - q(c+1)$$

$$\text{——————}(A)$$

Since $\Delta = mq - np \neq 0$, then

$$h = \frac{\Delta_1}{\Delta}, \qquad k = \frac{\Delta_2}{\Delta}$$



Example

- Solve $\left(x^{1/2}y - xy^2\right)dx + \left(x^{3/2} + x^2y\right)dy = 0.$
- Solution: Write the equation as
  $x^{1/2}(y\,dx + x\,dy) + xy(-y\,dx + x\,dy) = 0.$
- Comparing with (5), we have
  $a = \frac{1}{2}, b = 0, \quad c = 1, d = 1, m = 1, n = 1, \quad p = -1, q = 1.$
- $\therefore$ (A) reduces to

$$h - k = -\frac{1}{2}, \qquad h + k = -4$$

- Its solution is $h = -\frac{9}{4}, \quad k = -\frac{7}{4}.$
- Thus, IF is $x^{-9/4}y^{-7/4}$.
- The general solution of the DE is

$$\boxed{-\frac{4}{3}x^{-3/4}y^{-3/4} + 4x^{-1/4}y^{1/4} = K}$$ (HOMEWORK).



Linear DEs

- A **first-order linear differential equation** is an equation of the form (Known as **Lagrange Form**)

$$\frac{dy}{dx} + P(x)y = Q(x), \quad x \in I, \tag{6}$$

where $P(x)$ and $Q(x)$ are continuous functions on an interval I.

- If $Q(x) = 0$, the equation is called a **homogeneous linear differential equation**; otherwise, it is called a **non-homogeneous linear differential equation**.
- The linear DE can be written as $[P(x)y - Q(x)]\,dx + dy = 0$, so that we can take $M = P(x)y - Q(x)$, $N = 1$.
- Here $M_y = P(x)$ and $N_y = 0$, and hence the DE is not exact.
- Now

$$\frac{M_y - N_x}{N} = \frac{P(x) - 0}{1} = P(x) \quad \text{is a function of } x \text{ only.}$$



Linear DEs

- $\therefore$ The IF for a linear DE is $\mu(x) = e^{\int P(x)\,dx}$
- Multiplying (6) by the integrating factor,

$$\mu\frac{dy}{dx} + \mu P(x)y = \mu Q(x). \tag{7}$$

- Since

$$\frac{d}{dx}(\mu y) = \mu\frac{dy}{dx} + \mu P(x)y,$$

(7) becomes

$$\frac{d}{dx}(\mu y) = \mu Q(x). \tag{8}$$

- Integrating both sides of (8) gives the GS of the LDE as

$$\boxed{\mu y = \int \mu Q(x)\,dx + C,\ C \text{ is arbitrary constant.}}$$



Linear DEs

- Note that

$$\frac{M_y - N_x}{M} = \frac{P(x)}{P(x)y - Q(x)} \quad \text{is } \textit{not} \text{ a function of } y \text{ only.}$$

- Thus, $e^{-\int \frac{M_y - N_x}{M}\, dy}$ is **not** an IF of (6).
- A linear differential equation in x is given by

$$\frac{dx}{dy} + P(y)x = Q(y). \tag{9}$$

- Similarly, we will get IF of (9) as $\mu = e^{\int P\, dy}$.
- Thus, the GS of the LDE (9)

$$\boxed{x = e^{-\int P\, dy}\left[\int Q\, e^{\int P\, dy}\, dy + C\right]}$$



Examples

- Solve $\frac{dy}{dx} + 2y = e^{-2x}$.
- **Solution.** It is a LDE in y, and $P(x) = 2, \quad Q(x) = e^{-2x}$.
- The integrating factor is $\mu = e^{\int 2\,dx} = e^{2x}$.
- Thus, $e^{2x}\frac{dy}{dx} + 2e^{2x}y\frac{d}{dx}(e^{2x}y) = 1$ is exact DE.
- The general solution is $ye^{2x} = x + C$ or $y = e^{-2x}[x + C]$.
- Solve $\frac{dy}{dx} + \frac{2}{x+1}y = (x+1)^4, \quad x > -1$.
- Solve $\frac{dy}{dx} + \frac{2}{x}y = \frac{\sin x}{x^2}, \quad x > 0$.
- Solve $\frac{dy}{dx} + \tan x\, y = \sin x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$.
- Solve $(e^y - 2xy)\,y' = y^2$ (Linear in x)
- Solve $xy' + x = x^2(y-1)\,y'$ (Linear in x)
- Solve $y - xy' = y'y^2e^y$ (Linear in x)
- Solve $\sin y\,\dfrac{dy}{dx} = \cos x\,(2\cos y - \sin^2 x)$ (Put $\cos y = z$)
- Solve $x^3y' + 4x^2\tan y = e^x \sec y,\ y(1) = 0.$ (Put $\sin y = z$)



Bernoulli DE: Definition

- A first order DE of the form

$$y' + P(x)y = Q(x)y^n, \qquad n \neq 0,\ n \neq 1$$

  is called a **Bernoulli DE**.
- Note that $n = 0$ or $n = 1$ corresponds to a linear ODE.
- Let $v = y^{1-n}$, so $y = v^{\frac{1}{1-n}}$.
- Then $y' = \dfrac{1}{1-n} v^{\frac{1}{1-n}-1} v' = \dfrac{1}{1-n} v^{\frac{n}{1-n}} v'$.
- Substituting into the DE:

$$\frac{1}{1-n} v^{\frac{n}{1-n}} v' + P(x) v^{\frac{1}{1-n}} = Q(x)\, v^{\frac{n}{1-n}}$$

- Dividing by $v^{\frac{n}{1-n}}$ (assuming it is nonzero) gives a linear ODE in v.



Bernoulli DE: Reduction to Linear Form

- Dividing through:

$$\frac{1}{1-n}v' + P(x)\,v^{\frac{1-n}{1-n}-\frac{n}{1-n}} = Q(x)$$

- $$\Rightarrow \quad \frac{1}{1-n}v' + P(x)\,v = Q(x)$$

- $$\Rightarrow \quad \boxed{v' + (1-n)P(x)\,v = (1-n)Q(x)}$$

- This is of **linear Lagrange first order ODE** form.
- Solve for v, then recover $y = v^{\frac{1}{1-n}}$.



Example: $3y^2y' + y^3 = e^{-x}$

- Obtain the general solution of $3y^2y' + y^3 = e^{-x}$.
- Rewrite: $y' + \frac{1}{3}y = \frac{e^{-x}}{3}\, y^{-2}$, so $n = -2$ here.
- Take $v = y^{1-(-2)} = y^3$; then $v' = 3y^2y'$.
- The DE becomes $v' + v = e^{-x}$, a linear Lagrange first order ODE.
- I.F. $= e^{\int dx} = e^x$.
- $\therefore\ ve^x = \int e^x e^{-x}\, dx + C = x + C.$
- $\therefore\ v = \frac{1}{e^x}(x + C)$
- or $\quad y^3 = (x + C)e^{-x}$
- The General solution is

$$\boxed{y = (x + C)^{1/3} e^{-x/3}, \quad \forall x \in \mathbb{R}}$$



Example: Solve $y' = y(xy^3 - 1)$

- The given DE is $y' + y = xy^4$, which is a Bernoulli DE with $n = 4$.
- Take $v = y^{1-4} = y^{-3}$, so $y = v^{-1/3}$.
- $\dfrac{dy}{dx} = -\dfrac{1}{3} v^{-4/3} \dfrac{dv}{dx}$
- $\therefore \; -\dfrac{1}{3} v^{-4/3} \dfrac{dv}{dx} + v^{-1/3} = x\, v^{-4/3}$
- or $\dfrac{dv}{dx} - 3v = -3x$ (linear Lagrange first order ODE)
- IF$= e^{-\int 3\, dx} = e^{-3x}$
- The solution will take the form
$$ve^{-3x} = \int -3xe^{-3x}\, dx + C = \left(x + \frac{1}{3}\right) e^{-3x} + C$$
- The General solution is $\boxed{y = \left[x + \dfrac{1}{3} + Ce^{3x}\right]^{-1/3}}$



BDE - Problem

- Solve $x\dfrac{dy}{dx} + 3y = x^2y^{1/2}, \quad x > 0.$
- Write the equation as $\dfrac{dy}{dx} + \dfrac{3}{x}y = xy^{1/2}.$
- Here, $n = \dfrac{1}{2}.$
- Divide by $y^{1/2}$: $\quad y^{-1/2}\dfrac{dy}{dx} + \dfrac{3}{x}y^{1/2} = x.$
- Let $v = y^{1/2}$, so $\dfrac{dv}{dx} = \dfrac{1}{2}y^{-1/2}\dfrac{dy}{dx}.$
- Hence, $2\dfrac{dv}{dx} + \dfrac{3}{x}v = x.$
- or $\dfrac{dv}{dx} + \dfrac{3}{2x}v = \dfrac{x}{2}$, a linear DE in v.



Solving the Linear DE

- The integrating factor is $\mu = e^{\int \frac{3}{2x}\,dx} = x^{3/2}$.
- Therefore, $\dfrac{d}{dx}\left(x^{3/2}v\right) = \dfrac{1}{2}x^{5/2}$.
- Integrating, $x^{3/2}v = \dfrac{x^{7/2}}{7} + C$.
- Hence, $v = \dfrac{x^2}{7} + \dfrac{C}{x^{3/2}}$.
- Since $v = \sqrt{y}$, we obtain

$$\boxed{y = \left(\frac{x^2}{7} + \frac{C}{x^{3/2}}\right)^2}.$$

- This is the required general solution.

